STAT6611

- Lecture 19 (R, Oct 29)
- Solutions/grading key posted for midterm (exams to be posted on Canvas)
- Homework 4 assigned
  due Friday, Nov 13
  (practice on normal dist. & quadratic forms)

Lemma 3.6. *If A, B, and $A+B$ are all idempotent matrices, then $AB=0$.*

Proof. We have $A=A^2$, $B=B^2$, and

$$A+B=(A+B)^2=A^2+B^2+AB+BA=A+B+AB+BA,$$

implying that $AB=-BA$.

A, B idempotent

$(AB)^2 = ABAB = -A^2B^2 = -AB$

$= -AB$

$(-BA)^2 = BABA = -B^2A^2 = -BA$

$= -BA$

So, $AB = BA = -BA = 0$

Lemma 3.7. *Let $A_1,\ldots,A_k$ be $m\times n$ matrices. Then,*

$$\operatorname{rank}(A_1+\cdots+A_k)\le \operatorname{rank}(A_1)+\cdots+\operatorname{rank}(A_k).$$

Proof. By Lemma 2.2, we have

$$\operatorname{rank}\left(\sum_{i=1}^{k}A_i\right)=\operatorname{rank}\left([I_m,\ldots,I_m]\begin{bmatrix}A_1\\ \vdots\\ A_k\end{bmatrix}\right)\le \operatorname{rank}\begin{bmatrix}A_1\\ \vdots\\ A_k\end{bmatrix}\le\sum_{i=1}^{k}\operatorname{rank}(A_i).$$

Lemma 2.2

product of matrices

□

First we prove this $k=2$, $A_1 + A_2 = I - A_1$ by Lemma 3.5/3.6 above

Theorem 3.17. *Let $A_1, \ldots, A_k$ be $n \times n$ matrices such that $A_1 + \cdots + A_k = I_n$. Then, the three conditions are equivalent:*

(i) $A_i A_j = 0$ *for* $i = 1, \ldots, k$, $j = 1, \ldots, k$, *and* $i \neq j$;

(ii) $A_i = A_i^2$ *for* $i = 1, \ldots, k$;

(iii) $\sum_{i=1}^{k} \text{rank}(A_i) = n$.

Proof. It prove the theorem, it suffices to show that (i) $\Longrightarrow$ (ii), (ii) $\Longrightarrow$ (iii), and (iii) $\Longrightarrow$ (i).

Proof that (i) $\Longrightarrow$ (ii). If $A_i A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$, then

$$A_i = A_i I = A_i(A_1 + \cdots + A_k) = A_i^2 \quad \text{for} \quad i = 1, \ldots, k.$$

Proof that (ii) $\Longrightarrow$ (iii). If $A_i = A_i^2$ for $i = 1, \ldots, k$, then (by Lemma 2.5),

$$\sum_{i=1}^{k} \text{rank}(A_i) = \sum_{i=1}^{k} \text{tr}(A_i) = \text{tr}\left(\sum_{i=1}^{k} A_i\right) = \text{tr}(I) = n.$$

It is helpful to show that (iii) $\Longrightarrow$ (ii) before showing that (iii) $\Longrightarrow$ (i).

Proof that (iii) $\Longrightarrow$ (ii). If $\sum_{l=1}^{k} \text{rank}(A_l) = n$, then for $i = 1, \ldots, k$, by Lemma 3.7,

$$\text{rank}(I - A_i) = \text{rank}\left(\sum_{l=1}^{k} A_l - A_i\right) = \text{rank}\left(\sum_{l \neq i} A_l\right) \leq \sum_{l \neq i} \text{rank}(A_l) = n - \text{rank}(A_i).$$

Thus, $\text{rank}(A_i) + \text{rank}(I - A_i) \leq n$. By Lemma 3.7, we also have

$$\text{rank}(A_i) + \text{rank}(I - A_i) \geq \text{rank}(A_i + I - A_i) = \text{rank}(I) = n.$$

Thus, $\text{rank}(A_i) + \text{rank}(I - A_i) = n$. This implies that $A_i = A_i^2$ by Lemma 3.5.

Proof that (iii) $\Longrightarrow$ (i). If $\sum_{l=1}^{k} \text{rank}(A_l) = n$, then for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$, by Lemma 3.7,

$$\text{rank}(I - A_i - A_j) = \text{rank}\left(\sum_{l=1}^{k} A_l - A_i - A_j\right) = \text{rank}\left(\sum_{l \neq i, l \neq j} A_l\right)$$

$$\leq \sum_{l \neq i, l \neq j} \text{rank}(A_l) = n - \text{rank}(A_i) - \text{rank}(A_j).$$

Thus,

$$\text{rank}(A_i + A_j) + \text{rank}(I - A_i - A_j) \leq \text{rank}(A_i) + \text{rank}(A_j) + \text{rank}(I - A_i - A_j) \leq n.$$

By Lemma 3.7, we also have

$$\text{rank}(A_i + A_j) + \text{rank}(I - A_i - A_j) \geq \text{rank}(A_i + A_j + I - A_i - A_j) = \text{rank}(I) = n.$$

Thus, $\text{rank}(A_i + A_j) + \text{rank}(I - A_i - A_j) = n$, implying $A_i + A_j$ is idempotent by Lemma 3.5. Since (iii) $\Longrightarrow$ (ii), A_i and A_j are also idempotent. Then, by Lemma 3.6, $A_i A_j = 0$ for $i \neq j$.

Theorem 3.18. (**Cochran's Theorem**) *Let* $\underline{x} \sim N(\underline{\mu}, \sigma^2 I)$, *where* $\underline{x}$ *is an* $n \times 1$ *vector and* σ *is a positive scalar. Take* $A_1, \ldots, A_k$ *to be* $n \times n$ *symmetric matrices of ranks* $n_1, \ldots, n_k$ *such that* $A_1 + \cdots + A_k = I$. *(Thus,* $\underline{x}'\underline{x} = \sum_{i=1}^{k} \underline{x}' A_i \underline{x}$.*)*

If $\sum_{i=1}^{k} n_i = n$, *then* $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_1 \underline{x}, \ldots, \left(\frac{1}{\sigma^2}\right) \underline{x}' A_k \underline{x}$ *are mutually independent and* $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_i \underline{x} \sim \chi^2\left[n_i, \left(\frac{1}{2\sigma^2}\right) \underline{\mu}' A_i \underline{\mu}\right]$ *for* $i = 1, \ldots, k$.

Conversely, if $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_1 \underline{x}, \ldots, \left(\frac{1}{\sigma^2}\right) \underline{x}' A_k \underline{x}$ *are mutually independent and* $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_i \underline{x} \sim \chi^2(s_i, \lambda_i)$ *for some* s_i *and* λ_i *for* $i = 1, \ldots, k$, *then* $n_i = s_i$ *for* $i = 1, \ldots, k$ *and* $\sum_{i=1}^{k} n_i = n$.

Proof. Suppose that $\sum_{i=1}^{k} n_i = n$. Then, by Theorem 3.17, $A_i A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$, and $A_i = A_i^2$ for $i = 1, \ldots, k$.

Conversely, suppose that $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_1 \underline{x}, \ldots, \left(\frac{1}{\sigma^2}\right) \underline{x}' A_k \underline{x}$ are mutually independent and that $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_i \underline{x} \sim \chi^2(s_i, \lambda_i)$ for some s_i and λ_i for $i = 1, \ldots, k$.

by theorem 3.13. so $n = n_1 + \cdots + n_k = \sum_{i=1}^{k} \text{rank}(A_i)$

Cochran's theorem can be extended to the following result.

Theorem 3.19. *Let $\underline{x} \sim N(\underline{\mu}, \sigma^2 I)$, where $\underline{x}$ is an $n \times 1$ vector and σ is a positive scalar. Take $A_1, \ldots, A_k$ to be $n \times n$ symmetric matrices of ranks $n_1, \ldots, n_k$ such that $A_1 + \cdots + A_k = I$. (Thus, $\underline{x}'\underline{x} = \sum_{i=1}^{k} \underline{x}' A_i \underline{x}$.) Then, each of the following six conditions are equivalent:*

Theorem 3.17

(i) $A_i A_j = 0$ *for* $i = 1, \ldots, k$, $j = 1, \ldots, k$, *and* $i \neq j$;

(ii) $A_i = A_i^2$ *for* $i = 1, \ldots, k$;

(iii) $\sum_{i=1}^{k} n_i = n$;

Cochran's Theorem

(iv) $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_1 \underline{x}, \ldots, \left(\frac{1}{\sigma^2}\right) \underline{x}' A_k \underline{x}$ *are mutually independent;*

(v) $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_i \underline{x} \sim \chi^2 \left[n_i, \left(\frac{1}{2\sigma^2}\right) \underline{\mu}' A_i \underline{\mu}\right]$ *for* $i = 1, \ldots, k$;

just true

(vi) $\left(\frac{1}{\sigma^2}\right) \underline{x}' A_i \underline{x} \sim \chi^2 (s_i, \lambda_i)$ *for some* s_i *and* λ_i *for* $i = 1, \ldots, k$.

theorem 3.16

$\frac{\underline{x}' A_i \underline{x}}{\sigma^2}$ $\frac{\underline{x}' A_j \underline{x}}{\sigma^2}$ are independent

$\frac{A_i}{\sigma^2} \frac{\sum A_j}{\sigma^2} = 0$

since Σ is invertible

Proof. By Theorem 3.17, (i) $\iff$ (ii) $\iff$ (iii).

By Cochran's theorem, (iii) $\implies$ (iv).

By the remark after Theorem 3.14, (iv) $\implies$ (i).

Thus, (i) $\iff$ (ii) $\iff$ (iii) $\iff$ (iv).

By Cochran's theorem, (iii) $\implies$ (v).

Immediately, (v) $\implies$ (vi)

Theorem 3.16, (vi) $\implies$ (ii).

Therefore, each of the six conditions implies the other five.

□